

NOAA Passive Acoustic Monitoring Strategic Initiative (PAM SI): Workshop Synthesis Report

September 24 & 25, 2026



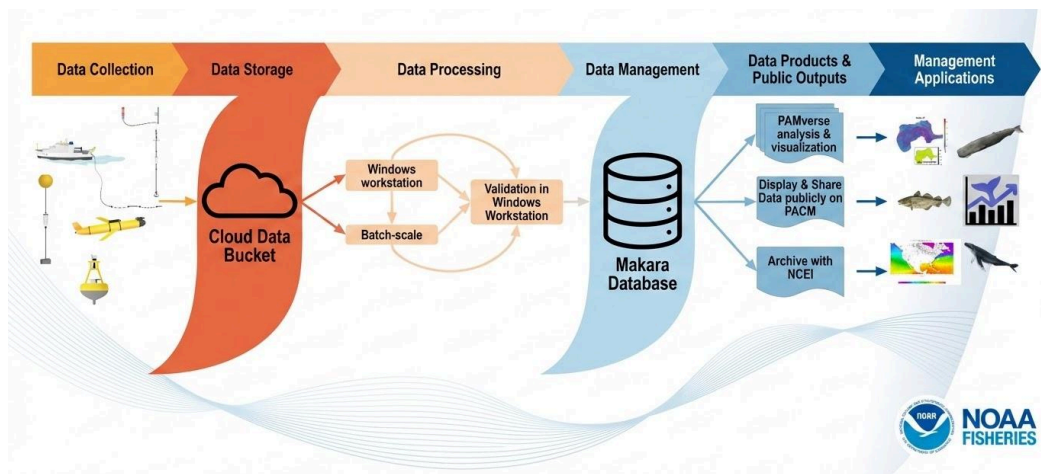
Executive Summary

The NOAA Passive Acoustic Monitoring Strategic Initiative (PAM SI) workshop brought together researchers, software engineers, and bioacousticians across NOAA Fisheries Science Centers and other NMFS, NOS and NCEI line offices, to evaluate progress across two core areas: **Day 1: PAM Analytical Developments** and **Day 2: National PAM Cloud Infrastructure**. 73 attendees attended day 1 and 65 attended day 2. The PAM SI has established an end-to-end, national enterprise-level ecosystem designed to transition passive acoustic data from isolated desktop analysis into scalable, cloud-native pipelines. Analytical developments have focused on integrating uncrewed platforms (gliders and drifters) and bottom mounted recorders into Species Distribution Models (SDMs), expanding fish PAM into formal stock assessments and standardizing national soundscape reporting. To date over 2.4 Petabytes of raw audio has been migrated to the shared PAM Google Cloud Platform (GCP). The development of a joint GCP infrastructure and shared batch scaling processes have exponentially sped up data analysis. The National Makara database represents a key achievement and is the only implemented NOAA Fisheries example of a centralized database for the agency. It is supported by pipeline developments that ease data input to Makara and egress to the PACM public webviewer, PAMverse analytical R Studio Packages and NCEI data archiving via the Tugboat process. The PAM SI has demonstrated drastically reduced analytical timelines, improved data transparency, and accelerated science-based marine data products.

PAM SI Working Group:

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Part I: Day 1 — PAM Analytical Developments



1. SPACIOUS: Integrating Slow-Moving PAM Platforms into Species Distribution Models

- Project Scope & Core Goals:** The SPACIOUS (Spatial Population Assessment of Cetaceans Integrating Acoustic Observations from Uncrewed Systems) PIFSC/SWFSC project develops analytical frameworks to incorporate passive acoustic data from slow-moving autonomous platforms—specifically Seagliders and Drifting Acoustic Spar Buoy Recorders (DASBRs)—into odontocete Species Distribution Models (SDMs) and spatial density estimations. The primary case study focuses on sperm whales (*Physeter macrocephalus*) across the U.S. West Coast (CalCurCEAS 2024) and Hawaii (WHICEAS 2026).

2. East Coast DASBRs: Expanding Acoustic Density Estimation

- Project Scope & Objectives:** The NEFSC Passive Acoustics Branch builds on previous PIFSC/SWFSC drifter projects and expands DASBR deployments to the northeast coast of the US across three field seasons (August 2025, May 2026, August 2026). The project targets deep-diving, cryptic species—specifically beaked whales—that are difficult to observe visually but produce frequent echolocation clicks while foraging at depth. Density estimates will be produced for beaked whales, spatial presence for other odontocetes and baleen whales, alongside ocean soundscape metrics, telemetry detections of tagged animals (e.g. fish, turtles and sharks) in addition to temperature data in support of oceanographic modeling.

3. Gulf LISTEN: Linking Ocean Dynamics to Cetacean Distributions

- Project Scope & Framework:** The Gulf LISTEN (Long-Term Investigations into Soundscapes, Trends, Ecosystems and Noise) SEFSC/Scripps project utilizes stationary High-frequency Acoustic Recording Packages (HARPs) across 21 monitoring sites (8 continuous long-term sites from 2020–2024 and 14 rotating 1-year sites). Operating in a heavily industrialized region, the study links 3D oceanographic drivers to deep-diving cetacean density. Species distribution models were developed for six deep-diving species

that are challenging to study using traditional visual survey methods. Models were developed using advanced machine-learning-based Boosted Regression Trees that integrate species density time-series from HARP recordings and 28 oceanographic condition predictors, including deep ocean circulation patterns. A robust model framework was developed for data resampling into training, testing, and evaluation datasets to enhance model generalization and performance. The study applied SHapley Additive exPlanations (SHAP) to evaluate predictor importance. Daily-resolution Gulf-wide species density maps were created by predicting species density over a 40 km² grid (the GoMMAPPS visual-survey-based distribution model grid) and resulting densities were summed over the Gulf to produce an abundance estimate timeseries. Methods are currently being developed to integrate visual-survey-based densities with moored-acoustic-based densities to improve abundance estimations, and resulting estimates are expected to be incorporated into beaked whale stock assessment reports (SARS).

4. PAM Applications for Fisheries Management and Stock Assessments (SARs)

- **Project Scope & Objectives:** A NEFSC/WHOI CINAR collaboration has worked on expanding capabilities for using known soniferous fish species calls through analysis of long term time series of PAM data initially in the NE. Seasonal and spatial patterns inform the presence and spawning activity of key species. As fish vocalizations have low source levels and short detection radii (tens of meters), fish PAM functions as localized point-sampling to complement trawl/visual surveys during spawning aggregations. A cross-division NEFSC and WHOI team are integrating PAM grunt rates into official Atlantic Cod and Haddock SARs. High potential exists for using PAM to support fisheries management and SARs across other regions.

5. U.S. Ocean Soundscape Monitoring & Condition Reporting

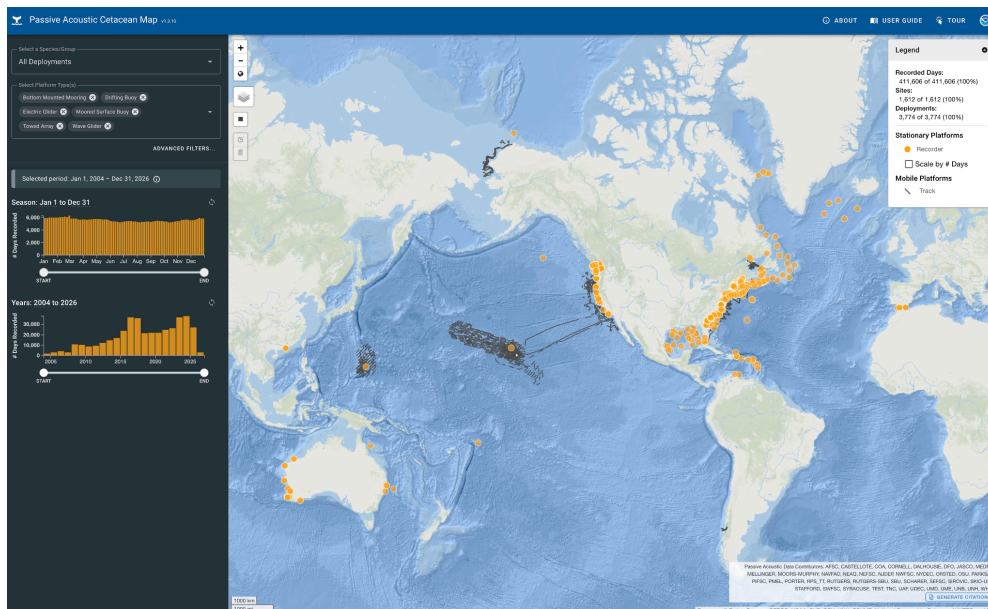
- **Framework & Core Goals:** The U.S. Ocean Sound Monitoring and Condition Reporting program establishes a standardized national framework that unifies the dissemination of soundscape data from the NOAA Office of National Marine Sanctuaries (ONMS) Sound Monitoring Program, the NOAA Ocean Noise Reference Station (NRS) Network, and NMFS Fisheries Science Centers. This coordinated effort and public-facing platform summarizes long-term acoustic datasets to permit examination of trends in ocean soundscapes, acoustic presence of indicator species, and measures of anthropogenic activity. Started as an ONMS effort, recent incorporation of NRS and NMFS datasets in coordination with the PAM SI has expanded this platform into a unique and data rich resource. ONMS focuses on tracking trends across Marine Sanctuaries, NRS provides long term deep and shallow water data records, and NMFS FMCs provide data coverage across diverse ecosystems throughout US waters.

6. PAMverse RStudio Ecosystem: Data Integration & Environmental Matching

- **New Core Functionalities:** The PAMverse ecosystem provides open-source R tools to query cloud databases, integrate spatio-temporal data, and standardize archive submissions. Accessing centralized detection metadata in Makara is executed within the NOAA Cloud Workstation (pam-ww) via Google BigQuery. Environmental Matching (matchEnvData) automatically appends spatio-temporal environmental variables to any R dataframe containing UTC, Latitude, and Longitude columns. Works seamlessly with loadSoundscapeData, which extracts spatial coordinates directly from NetCDF metadata.

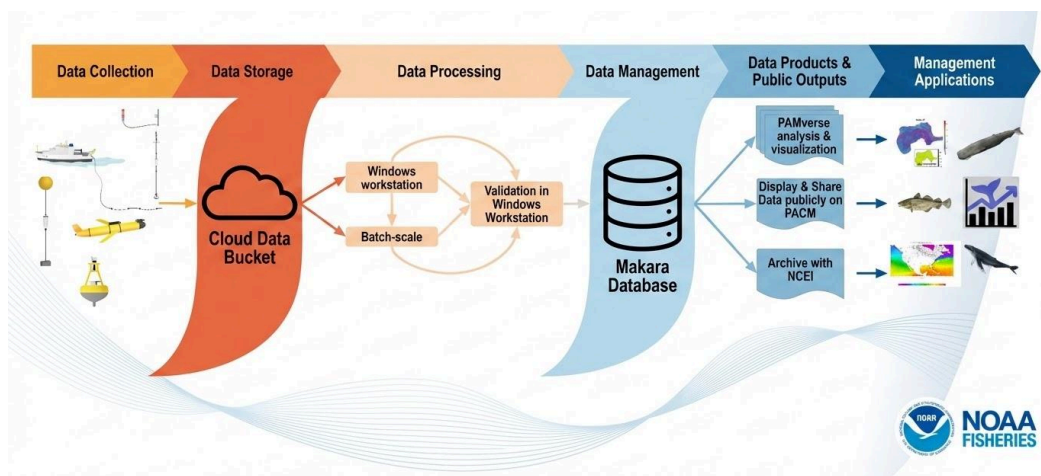
7. Makara Database, PACM, and PARS Data Ecosystem

- **Enterprise Database Architecture:** The **Makara Passive Acoustics Database** serves as the central data management hub for NOAA Fisheries. Built on Google Cloud Platform (Google Cloud SQL PostgreSQL engine and Google BigQuery), Makara links cloud storage assets, analytical scripts (**PAMverse Rstudio**), visualization portals (**PACM**), and external partner submission forms (**PARS**).
- **Portal Suite:**
 - **Makara Frontend:** User interface for account management, table editing, batch CSV template importing/exporting, and BigQuery downloads.
 - **Passive Acoustic Cetacean Map (PACM):** Public interactive map displaying deployment metadata, grouped monitoring sites, and detection time-series with automated citations. Contributions across FMCs are rapidly growing.



- **Passive Acoustic Reporting System (PARS):** Public submission portal for non-NOAA partners (universities, industry, NGOs) to contribute acoustic detections.

Part II: Day 2 — National PAM Cloud Work



8. National PAM Cloud Overview & Modernized Batch Workflows

- Legacy Challenges vs. Cloud Solution:** Legacy PAM workflows relied on local Science Center network drives and external hard drives, causing severe analytical bottlenecks where sequential desktop processing across 50 recorders took up to 150 days. The National PAM Cloud Initiative migrates on-premise PAM workflows to elastic, parallel cloud computing on GCP, hyper-scaling processing while providing analysts with a virtual Windows interface.
- Parallel Batch Scaling & Virtual Workstations:** Replacing sequential desktop execution with Google Cloud Batch clusters achieves up to a **95% reduction in processing time** (reducing multi-month runs to 3 days). The Windows Virtual Machine (*pam-ww*), developed with NMFS OCIO, provides a familiar desktop environment for interactive data validation and spectrogram inspection directly in the cloud.
- Orchestration & Machine Learning Models:** Google Cloud Composer serves as a centralized UI to simultaneously run over 10 specialized acoustic detection pipelines. Automated processing is carried out using the Google Multi-Species Whale ML Model, IDL LFDSCS baleen whale detector, INSTINCT and PyPAM HMD soundscape processing. Complete open-source documentation is available on GitHub (<https://nmfs-ost.github.io/PAM-Cloud/>)(<https://nmfs-ost.github.io/PAM-Cloud/>)).

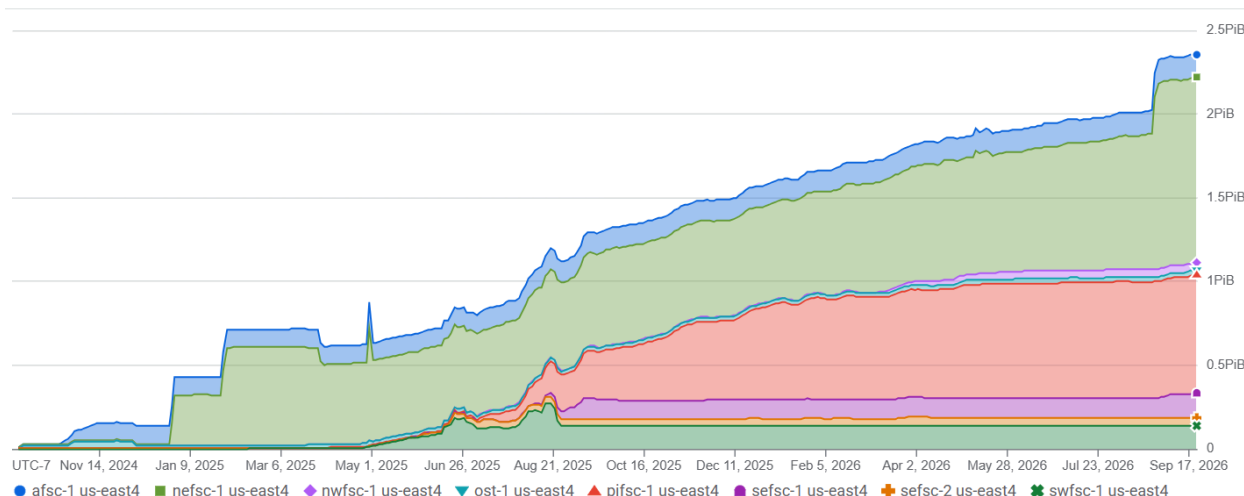
9. NCEI Archiving Pipeline & Tugboat

- Archiving Scale & Traditional Bottlenecks:** NOAA Fisheries manages over **3 Petabytes** of PAM data across science centers. Legacy archiving relied on manual hard drive staging and physical shipping (FedEx).
- Tugboat Cloud Ingest Service:** NOAA NCEI/Fisheries collaboration. **Tugboat** is a cloud-native job orchestration application that automates, validates, and expedites transferring big acoustic datasets from cloud storage into the NCEI Ocean Acoustic Archive. Tugboat is cloud-provider agnostic, pulling data from GCP or Azure buckets using service accounts restricted strictly to read-only permissions. Users trigger archive

jobs via the Makara frontend ([Archive to NCEI](#)). Makara validates required metadata fields against Tugboat API rules, transforms the metadata into a submission package, and tracks job status ([PENDING](#) → [RUNNING](#) → [COMPLETED](#)).

10. Cloud Storage Architecture, Ingest, and Cost Optimization

- **Cloud Footprint & Data Scale:** NOAA Fisheries has migrated over **2.407 PiB (72.748 million objects)** out of **~3.5 PiB** of acoustic data across 49 cloud buckets on GCP (NEFSC ~1.3 PiB, PIFSC ~650 TB, SWFSC & SEFSC & AFSC ~150 TB).



- **Ingest & Server-Side File Operations:** Data are uploaded via command-line tools ([rclone](#), [gcloud storage](#)), custom GUIs (Tony Yang Data Transfer Tool), or Google Storage Transfer Appliances. Because GCS object storage is atomic (renaming/moving a file is a create/delete operation), moves must be executed server-side ([gcloud storage mv](#), [rclone move](#), custom GCS Move/Rename GUI) to avoid downloading data locally. Analysts should avoid drag-and-dropping files over GCS mounts for performance reasons, and when not on a cloud compute instance, high costs.
- **Cost Structure & Autoclass Analysis:** GCP storage tiers span Standard, Nearline, Coldline, and Archive. Buckets currently operate under **Autoclass**, which automatically transitions untouched files to Nearline after 30 days. Semi-annual cost audits demonstrate that Autoclass is optimal against hypothetical alternate storage class settings overall because of out use patterns. Forcing colder tiers upfront introduces early deletion penalties and access cost spikes when datasets experience higher use.

11. Cloud-Based Navy Review & Data Handling Protocols

- **Context & Security Mandate:** NOAA Fisheries PAM operations may occasionally record sensitive acoustic signals in areas with national security implications which led to a NMFS/Navy agreement allowing Navy to screen potentially sensitive datasets. Recently, the U.S.Navy expanded their requests to include approximately 40% of NOAA Fisheries' PAM datasets. The renewal of the agreement, which expired in 2025, is

currently under negotiation considering the large volume of data requested for screening. NMFS OST is coordinating across FMCs to preserve research access and establish a cloud-native screening pipeline.

- **FedRAMP High GCP Landing Zone:** NOAA OCIO deployed a FedRAMP High GCP Landing Zone environment (September 2026) capable of hosting FISMA High, Moderate, or Low systems. Security leads are completing risk assessments to secure an Authority to Operate (ATO) or Authority to Use (ATU), allowing the Navy to approve the cloud environment for potentially sensitive data.
- **Cloud Pre-Clearance Pipeline:** Recovered PAM audio will be uploaded directly into a Pre-clearance Cloud Bucket with restricted access. Once Navy personnel screen and clear/redact the data, audio moves to Cleared/Redacted Buckets to trigger Makara metadata validation, PACM mapping, and NCEI archiving. Planned and completed deployment metadata entered into Makara will automatically populate a SmartSheet database to alert Navy Undersea Surveillance (CUS).

12. Cloud Windows Workstations ([pam-ww](#))

- **Architecture:** The PAM Windows Workstation ([pam-ww](#)) provides a cloud-hosted virtual desktop environment on GCP, allowing analysts to run desktop software (Raven, Triton, PAMGuard) directly adjacent to cloud storage. The fleet currently supports **75 active virtual machine instances** across all NOAA Fisheries Science Centers. Hardware profiles include standard ([pam-ww](#)), high-compute ([pam-ww-extension](#)), and GPU-accelerated ([pam-ww-gpu](#)) instances.
- **Cloud Mounts & Read-Ahead Caching:** Direct virtual disk mounts map GCP storage buckets into Windows File Explorer. A custom-commissioned user mount and read-ahead cache software buffers acoustic audio files locally to accelerate file retrieval during visual/auditory analysis. IDL integrated license server is supported
- **Technical Challenges & Cost Structure:** Cloud file reading is functionally a network download, introducing "time-to-first-byte" latency when opening small files. Operational costs are overwhelmingly dominated by **Windows Server 2022 Datacenter OS licensing fees** rather than compute usage.

13. Cloud Batch-Scale Processing & Workflow Demonstrations

- **Orchestration via Airflow DAGs:** Live demonstrations showcased Google Cloud Composer (Apache Airflow) running automated DAGs. The multi-user architecture allows multiple analysts to trigger processing jobs concurrently, tracking job parameters and execution logs individually.
- **Demonstrated Pipelines:**
 1. **Google Multi-Species ML Model DAG:** Runs Google's 8-class classifier across cloud storage buckets, exporting detection scores to BigQuery.
 2. **SoundChecker Spectrogram Pre-Generation ([wav2png](#)):** Converts raw audio into pre-rendered PNG spectrograms ([wav2png_cloud_batch_orchestrator](#)) to support rapid manual paging in SoundChecker.

3. **MATLAB Containerization:** Compiles legacy MATLAB bioacoustic routines into executable Docker containers running inside Google Cloud Batch, eliminating software license bottlenecks.
4. **LFDCS Cloud Pipeline:** Launches LFDCS by providing a Makara deployment code to auto-generate parameter files and trigger processing via the launcher DAG.

14. Annotation to AI Classification (Perch / Hoplite)

- **Core Innovation & Agile Modeling:** Traditional bioacoustic machine learning ML requires thousands of manual annotations and complex re-training cycles. NOAA Fisheries partnered with Google Research to implement **Agile Modeling** via **Perch Hoplite**. By using a frozen audio foundation model (Perch 2.0) to convert raw audio into vector embeddings, analysts build classifiers using **few-shot active learning**—requiring only a few call examples without writing ML code.
- **Two-Step Architecture:**
 1. **Step 1: Embedding Generation (One-Time Execution):** Raw audio is passed through a foundation model (default Perch 2.0) to extract vector embeddings (5-second audio "fingerprints") via an Airflow DAG (`perch_hoplite_embedding_job_dag`). Executed once per dataset, vectors are stored permanently in a **LanceDB** database on GCP.
 2. **Step 2: Few-Shot Active Learning (Hoplite GUI):** An analyst provides a few call examples in an intuitive GUI. Hoplite rapidly searches LanceDB embeddings for vectors with similar features, presents similar spectrogram results, receives analyst feedback, and quickly trains a custom classifier. Several rounds of active training help refine the classifier before it is scaled across multi-year datasets.
- **Database Schema Integration:** Makara handles deployment provenance and daily presence summaries. A dedicated **Annotation DB** manages millions of individual call-level annotations, LanceDB stores embedding vectors, and BigQuery houses raw inference results.

Part III: PAM Useful links

Workshop agenda, notes and presentations

[PAM SI Meetings](#)

For Makara Database, PACM webviewer and PARS external collaborator data submissions:

[National PAM SI Database and Viewing Applications](#)

For PAM Cloud SOPs and viewing all PAM SI projects:

[National PAM SI Cloud SOPs](#)

For PAM analyses and integrative data products:

[PAMverse R Studio Packages](#)

The PAMverse consists of three R packages:

- [PAMscapes](#): Functionality includes the QAQC tools as well as the analytical and visualization functions for soundscapes, acoustic detections, and integrative products. Most of the recent developments from the PAM SI are included here.
- [PAMmisc](#): A collection of helper functions that support the other packages including loading environmental data that aligns with a given dataset.
- [PAMpal](#): A collection of functions that work directly with PAMGuard data.

For NOAA NOS, NRS and NMFS sound metric reporting:

[U.S. Ocean Sound Monitoring and Condition Reporting](#)